

AOC-0 / R-2 Flight operations manual

Orbital procedures and checklists



About this document

This document describes the procedures to be used during a AOC-0 orbital mission. This mission requires an R-2 launcher and all known mission variants are described in this document. This document also contains the checklists and actions to be taken under unusual and emergency circumstances.

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Mission types

There are 2 known mission types for the AOC-0 in a orbital configuration: exo-atmospheric exploration and exo-atmospheric tourism. Tourism missions are crewed by an unqualified kerbal and are only performed in equatorial orbits while exploration missions require an able crewman and can be performed on different inclinations. The missions are all performed using an R-2 800 rocket.

The missions are assumed to happen with limited support and communications from ground. Because of that, tourism missions are forbidden for everything but equatorial inclinations.

Final assembly

Mount AOC-0 capsule on top of SM-1 decoupler.

Parachutes should align to R-2 parachutes

Hatch should align with R-2 gas generator exhaust

Hatch should align with SM-1 fin

Orient capsule/booster stack with capsule hatch pointing away of desired direction of gravity turn

Re-root to Mk1 Command pod

Staging should release lower stage first and upper stage last.

Stage	Parts	Notes
0	US Parachutes, US Decoupler, AOC Parachutes	
1	US Engine, LS Parachutes, LS decoupler	
2	LS Engine	

Quality control steps

If the decoupler has been separated from the booster, ensure that it is reconnected to the booster and capsule. If the decoupler is connected to the heat shield and the booster is then connected to the capsule, it is possible that the booster ends up connected directly to the heat shield via the second node in the heat shield, making it impossible to separate. This can be seen when the booster causes the shroud to appear around the heat shield despite the decoupler being connected to the heat shield.

Quick turnaround procedure

After an orbital mission and recovery of booster and capsule.

1. Mate capsule, service module and booster
2. Charge capsule battery
3. Charge O2 tank to 20%
4. Install fins on service module
5. Enable decoupler staging on both booster and service module

6. Charge propellants on booster and service module
7. If capsule batteries were drained, recharge computer battery.
8. Change staging to match launch configuration.

Orbital AOC-0 launch profile

The profile is a vertical launch with gravity turn to achieve a 100x100 km orbit. The lower stage is recovered after launch with the partially depleted upper stage remaining with the craft as its service module. The SM-1 is detached only after the re-entry burn and is recovered in a separate operation.

Flight parameters

Launcher R-2

Orbit 100Km circular

See launch section for details on the pitch over maneuver.

Equipment loadout

Resource	Amount	Units	Notes
Electrical charge	310	EC	RT value, fill all batteries to capacity
Monopropellant	0	%	No RCS to use monoprop
Spare parts	5	Units	
Food	EMPTY	%	LS is not an issue for short flights
Water	EMPTY	%	LS is not an issue for short flights
Oxygen	20	%	
Carbon Dioxide	EMPTY	%	
Waste water	EMPTY	%	
Waste	EMPTY	%	

Pilot should carry a parachute and EVA jetpack in exploration missions. Tourism missions do not carry equipment (they actually do, but cannot use them).

Food and water is not necessary for the short flights of the AOC-0. The limiting factor is EC supply and oxygen. Enough oxygen is provided to last at least 1 hour past the battery expiration time. The craft endurance is limited to 2:25 hours until battery is drained.

When the vessel is waiting recovery the limit is 18 hours because of lack of water. Recovery should be performed within that time.

Pre-rollout check

- Verify equipment loadout
 - Spare parts: 5
 - No monopropellant
 - Food: EMPTY
 - Water: EMPTY

- Oxygen: 22.2 (20%, 3h 36m)
- Waste/Wastewater/CO2: EMPTY
- Pilot carries parachute and EVA jetpack, tourists do not
- Verify R-2 propellant levels
 - Fully loaded
- Verify R-2 parachute parameters
 - Pressure 0.45
 - Altitude 500
 - Angle 10
 - Open when safe
- Verify AOC-0 parachute parameters
 - Pressure 0.6
 - Altitude 750
 - Angle 10
 - Open when safe
- Verify electrical system
 - Fully charged, both in Stayputnik and Mk1 pod
 - EC flow set to 0 in Mk1, -1 in Stayputnik, priority goes to Mk1
- Verify that booster decouplers are stageable
- Verify that no Dangit reports indicate damage
- Verify staging
 - LS engines on stage 2
 - US engines, LS decoupler, LS parachutes on stage 1
 - US parachutes, AOC-0 parachutes and US decoupler on stage 0

In flight checklists and procedures

SYSTEM LIST (CAPSULE)

Pod

Reaction wheel

Lights

Battery

Part storage bin

Food storage

Water tank

O2 tank

CO2 tank

Waste tank

Wastewater tank

Computer

Battery

Parachute (L)

Parachute (R)

Radio (L)

Radio (R)

Thermometer

Barometer

Heat shield

Shield

Ejector

SYSTEM LIST (BOOSTER)

LS Engine

LS Fins

LS LFOX tanks

LS Parachutes

LS decoupler

SYSTEM LIST (SERVICE MODULE)

US Engine

US Fins

US LFOX tanks

US Parachutes

US decoupler

MINIMUM EQUIPMENT LIST

Items marked with * are mandatory for tourism flights only

Capsule

Pod

Wheel

Battery

O2 tank

Computer

Computer Battery*

Parachute x2

Radio x2*/x1

Service module

Engine

Fins

LFOX tanks

Parachutes

Decoupler

Booster

Engine

Fins

LFOX tanks

Parachutes

Decoupler

NORMAL PROCEDURES

Pre flight

Computer

Hibernate OFF

Hibernate in warp AUTO

Battery FULLY CHARGED

Capsule

Parachutes	0.6/750/10/Disarmed
Battery	FULLY CHARGED
Spare parts	5
Food	EMPTY
Water	EMPTY
Oxygen	20%
Waste	EMPTY
Wastewater	EMPTY
Carbon Dioxide	EMPTY
Reaction wheels	NORMAL/ENABLED/IDLE
Radios	BOTH ENABLED

Service module

Parachutes	0.45/500/10/Disarmed
LFOX tanks	FULL
Fins	INSTALLED
Decoupler	STAGEABLE

Booster

Parachutes	0.45/500/10/Disarmed
Decoupler	STAGEABLE
LFOX tanks	FULL
Fins	ANGLE 0/LIMITER 4

Fin check

Capsule toggle torque WHEELS DISABLED
Test flight controls CHECK BOOSTER FINS
Capsule toggle torque WHEELS ENABLED

Check staging

Booster engine on stage 2

SM engine, booster parachutes and decoupler on stage 1

SM parachutes, AOC parachutes and decoupler on stage 0

Enable FMRS (if installed) to allow recovery of booster.

Launch

Set KER to orbit.

Stage to start engine

Rotate rocket to face to pitch over heading

Wait until pitch-over-maneuver speed is reached

Pitch over towards indicated heading until velocity vector reaches the indicated angle from vertical

Release controls after angle is reached

Profile	Launch point	POM speed (m/s)	POM heading (deg true)	POM angle (deg fm vert)	Max dV at orbit (m/s, expected)
Equatorial	KSC	50	090	30	370
Polar	KSC	50	180	25	140
Equatorial	Dessert	50	090	30	350
Polar	Dessert	50	000	25	150
Polar	Dessert	50	180	25	
Polar	Woomerang	50	000	25	180

Correct for any inclination errors at this point

Note that corrections require an expenditure of dV

Stage when booster propellants are exhausted

Continue burn using service module until apoapsis reaches 100 km, the cut thrust

For tourism missions, the decision point takes place right before the orbital insertion burn. By this point, the insertion burn must be programmed or the ship must be staged to allow for a safe launch abort (AST mode). When in doubt, perform the abort.

If using RT, program circularisation burn at apoapsis during coast maneuvering

This burn takes place out of line of sight from the KSC

If no trained pilot is on board, it is necessary to complete this before losing contact

Execute orbital insertion burn at apoapsis. Do not execute burn if remaining dV is estimated to be less than 30 m/s.

Orbital operations

There are no specific operations to execute. Verify resource status and initiate re-entry if EC or oxygen levels become too low.

Re-entry

KER

ORBIT/HEAT.

Attitude

ORBIT RETROGRADE

Check potential landing site and perform burn at the point indicated in the table below. Assume ± 2 degrees of error.

	Burn (m/s)	Angle of landing (degrees)
50		
100		92
150		72
200		60
250		54
300		48

Once retro burn is completely

Service module STAGE

If separation is unsuccessful,

Landing

Landing is automatic via parachute. No need to take any action unless there is a failure. Gross failures require pilot ejection.

EMERGENCY PROCEDURES

These procedures should be followed in case of unexpected events or damage to the spacecraft and are common to all suborbital operation modes.

Abort modes

Early launch abort (ELA)

This abort takes place in case of failures during the boost phase where the apoapsis is below 24 km. Thrust is cut, stages separate and land near the launching space port.

Booster Engine Thrust	CUT
-----------------------	-----

Booster	STAGE
---------	-------

If terrain avoidance is needed apply thrust with the SM-1. Cut thrust when avoidance is complete.

SM-1 Engine Thrust	CUT
--------------------	-----

SM-1	STAGE
------	-------

Land on parachutes downrange of launch point

Abort to suborbital trajectory (AST)

This abort mode happens between booster separation and the circularisation burn. Apoapsis is between 24 and 100 km. Thrust is cut, stages separate and capsule and service module land at the end of the suborbital trajectory defined by the partial launch.

SM-1 Engine Thrust	CUT
--------------------	-----

SM-1	STAGE
------	-------

If on a tourism flight, the staging should be done as early as possible while within line of sight of KSC.

Land on parachutes at the end of the suborbital trajectory.

Abort to extended suborbital trajectory (ASTE)

This abort mode occurs during the early parts of the orbital insertion burn, while the periapsis is below Kerbin's surface. This is conceptually identical to AST, with a landing point much downrange.

SM-1 Engine Thrust	CUT
--------------------	-----

SM-1	STAGE
------	-------

Land on parachutes at the end of the suborbital trajectory.

This abort mode is not possible on capsules without able crews (uncrewed or tourism) and no satellite communications.

Abort once around (AOA)

This abort mode happens during the orbital insertion burn with the periapsis above the surface but within the atmosphere. Care is taken not to propel the capsule into orbit during the abort. The capsule skips on the atmosphere and lands after speed has bled off.

SM-1 Engine Thrust CUT

SM-1 STAGE

Land on parachutes at the end of the suborbital trajectory.

This abort mode is not possible on capsules without able crews (uncrewed or tourism) and no satellite communications.

Abort once around with retroboost (AOA-R)

This abort mode happens when ATO would take place but is undesirable.

SM-1 Engine Thrust IMMEDIATELY CUT

Attitude ORBIT RETROGRADE

SM-1 Engine Thrust MAX

Position periapsis at least below 65 km altitude, then

SM-1 STAGE

Skip atmosphere and land at the end of suborbital trajectory.

Abort to orbit (ATO)

This abort mode occurs during orbital insertion burn with the apoapsis above Kerbin's atmosphere. In this mode, an alternate orbit is established and a second burn is executed at apoapsis to reduce the periapsis and deorbit by bleeding speed in the upper atmosphere.

SM-1 Engine Thrust CUT

Wait until periapsis is reached and a convenient landing point is found.

Attitude RETROGRADE

SM-1 Engine Thrust MAX

Execute de-orbit burn

SM-1 STAGE

Land on parachutes at the end of the re-entry trajectory.

This abort mode is not possible on capsules without able crews (uncrewed or tourism) and no satellite communications. A variant where the retro burn is executed over the KSC can be achieved for uncrewed capsules.

Attached re-entry abort, boost phase (ARA-B)

Thrust MAX

Wait until apoapsis is above 30km

Thrust	CUT
Parachute altitude (all)	MAX
Parachute pressure (all)	0.01
Parachutes (all)	DEPLOY

At apoapsis

Attitude	RETROGRADE
Thrust	MAX

Bleed as much speed as possible, maintain speed below 350 m/s. Parachutes deploy at 18 km altitude. Prevent nose down attitude at all costs.

Attached re-entry abort from orbit (ARA-O)

This re-entry mode is used when a capsule cannot or should not detach from the service module

Attitude	RETROGRADE
SM engine	BURN until PE is 50 Km
Parachute pressure (all)	MIN
Parachute altitude (all)	MAX
Parachute mode (all)	RISKY
Parachutes (all)	MANUAL DEPLOY

Do not activate TD-12 decoupler

If battery power is high (>50%)

Fly through 55 to 45 Km band of the atmosphere

Start gliding at 65 km

Optimal glide AOA is 40 degrees

Attempt to minimise vertical velocity

Bleed excess horizontal velocity

When battery power is low (50%) or craft is forced to drop below 45Km, stop flying

Attitude	RETROGRADE
----------	------------

Maintain retrograde attitude for as long as possible

At 45Km altitude

Throttle	MAXIMUM
----------	---------

Execute a retrograde burn to bleed excess speed and lose excess mass

If fins have not been destroyed at 40km altitude, destroy them by putting them through the plasma jet.

When forced to depart from retrograde attitude, actively attempt to maintain attitude as close to retrograde as possible

When parachutes deploy

Attitude CEASE CONTROL INPUTS

Capsule and service module land automatically attached to each other

Crew parachute re-entry (CPRE)

Remain in capsule until atmospheric re-entry is complete.

Wait until 9 Km altitude and 280 m/s speed is reached.

Perform EVA. Crewman should be dragged by airflow.

When terminal velocity of 60 m/s is reached, deploy parachute. Parachute should open instantly.

Configure parachute to 2500 m full opening.

When parachute opens, glide towards rescue point. Vertical speed should be 3.5 m/s and 20 m/s horizontal speed.

At 5 m above terrain, flare parachute to lose horizontal speed to less than 10 m/s and land.

During lower stage boost

Module	System failure	Abort mode	Notes	Check below
Capsule	Reaction wheel	ELA, immediate		No
	Battery	ELA, immediate		No
	Food leak	NOT PRESENT		No
	Water leak	NOT PRESENT		No
	O2 leak	ELA/AST/ASTE/ AOA	As needed to reach convenient landing point	No
	Computer Battery	ELA, immediate	Tourism only	No
	Radio x2	ELA/AST	Pilot decision	No
	Radio	ELA, immediate	Tourism only	No
	Parachute x1	ELA/AST/ARA-B	Depending on chance of second parachute failure	Yes
	Parachute x2	ARA-B		No
SM-1	Engine	AST	After LS separation	No
	Engine coolant	AST	After LS separation	No
	Engine gimbal	AST	After LS separation	No
	Fins	Special		Yes
	LFOX leak	AST	After LS separation	No
	Parachute x1	Ignore		No
	Parachute x2	Ignore	SM-1 will be destroyed on landing	No
	Decoupler	Special		Yes

Module	System failure	Abort mode	Notes	Check below
RS-1	Engine	Special		Yes
	Engine coolant	Special		Yes
	Engine gimbal	NOT PRESENT		No
	Fins (During POM) ELA			No
	Fins (Locked off centre)	ELA		No
	Fins (inert, stowed)	Ignore, continue burn	Should not be causing problems	No
	LFOX leak	Special		Yes
	Parachute x1	Ignore		No
	Parachute x2	Ignore	LS destruction	No
	Decoupler	Special		Yes

Pod parachute failure (single)

Execute ELA or ARA-B immediately. Select based on predicted failure of second parachute (ARA-B if failure likely, ELA otherwise).

Service module fins failure

This should only happen if fins were not installed during turnaround.

If possible, proceed with flight using pilot SAS.

Execute AST if vessel is uncontrollable.

Service module decoupler failure

Exploration mission may continue if decoupler can be repaired in orbit or heat shield jettison can be expected to work during re-entry. If in doubt or in tourism mission, perform procedure below.

Parachute state (capsule) DEPLOYED

Parachute state (service module) DEPLOYED

US Decoupler MANUAL DECOUPLE

If successful execute AST, else

Heat shield MANUAL JETTISON

If separation successful execute AST, else execute ARA-B procedure

Booster LFOX tank leak

See General booster engine failure section.

General booster engine failure (all modes including LFOX exhaustion)

Attempt to continue burn and monitor apoapsis. When in doubt or peril, execute ELA immediately.

Profile	Launch base	Threshold
Equatorial	KSC	20Km

Profile	Launch base	Threshold
If apoapsis is below critical threshold or uncertain Execute ELA, else		
Staging	ACTIVATE	
Proceed with burn using upper stage. Monitor dV and perform appropriate abort mode if propellants are insufficient.		

Booster decoupler failure

Thrust	CUT
LS Decoupler	MANUAL DECOUPLE
If successful, continue with launch, else	
Parachute state (capsule)	DEPLOYED
Parachute state (service module)	DEPLOYED
Parachute state (booster)	DEPLOYED
LS decoupler	MANUAL STAGE
IF successful Execute AST, else	
US Decoupler	MANUAL STAGE
IF successful Execute AST, else	
Heat shield	MANUAL JETTISON
IF successful Execute AST, else	
Thrust	MAX
US engine	MANUAL ACTIVATE
If separation successful execute AST, else	
Hatch	OPEN
Execute EVA and eject from craft	
Landing with full stack is not survivable.	

During upper stage boost

Module	System failure	Abort mode	Notes	Check below
Capsule	Reaction wheel	AST		No
	Battery	AST		No
	Food leak	NOT PRESENT		No
	Water leak	NOT PRESENT		No
	O2 leak	AST/ASTE/AOA	As needed to reach convenient landing point	No
	Computer Battery	AST, immediate	Tourism only	No
	Radio x2	AST	Pilot decision	No

Module	System failure	Abort mode	Notes	Check below
SM-1	Radio	AST, immediate	Tourism only	No
	Parachute x1	AST/ARA-B	Depending on chance of second parachute failure	Yes
	Parachute x2	ARA-B		No
	Engine	AST		No
	Engine coolant	AST		No
	Engine gimbal	AST		No
	Fins	Special		Yes
	LFOX leak	AST		No
	Parachute x1	Ignore		No
	Parachute x2	Ignore	SM-1 will be destroyed on landing	No
	Decoupler	Special		Yes

Pod parachute failure (single)

Execute AST or ARA-B immediately. Select based on predicted failure of second parachute (ARA-B if failure likely, AST otherwise).

Service module fins failure

This should only happen if fins were not installed during turnaround.

If possible, proceed with flight using pilot SAS.

Execute AST if vessel is uncontrollable.

Service module decoupler failure

Exploration mission may continue if decoupler can be repaired in orbit or heat shield jettison can be expected to work during re-entry. If in doubt or in tourism mission, perform procedure below.

Parachute state (capsule) DEPLOYED

Parachute state (service module) DEPLOYED

US Decoupler MANUAL DECOUPLE

If successful execute AST, else

Heat shield MANUAL JETTISON

If separation successful execute AST, else execute ARA-B procedure

During orbital insertion burn

Module	System failure	Abort mode	Notes	Check below
Capsule	Reaction wheel	ASTE/AOA AOA-R		Yes
	Battery	ASTE/AOA AOA-R		No
	Food leak	NOT PRESENT		No
	Water leak	NOT PRESENT		No

Module	System failure	Abort mode	Notes	Check below
SM-1	O2 leak	ASTE/AOA/ATO	As needed to reach convenient landing point	No
	Computer Battery	ASTE/AOA AOA-R	Tourism only	No
	Radio x2	ASTE/AOA/ATO	Pilot decision	No
	Radio	ASTE/AOA AOA-R Immediate	Tourism only	No
	Parachute x1	Special	Depending on chance of second parachute failure	Yes
	Parachute x2	ARA-O		No
	Engine	ASTE/AOA		No
	Engine coolant	ASTE/AOA AOA-R		No
	Engine gimbal	ATO		No
	Fins	N/A		No
	LFOX leak	AOA/AOA-R		No
	Parachute x1	Ignore		No
	Parachute x2	Ignore	SM-1 will be destroyed on landing	No
	Decoupler	Special		Yes

Reaction wheel failure

Use engine gimbal to attain retro attitude. Perform earliest abort possible.

Capsule parachute failure

Decide if second parachute failure is imminent.

If repairs in orbit are possible continue with mission and repair.

If repairs are not possible

If second failure imminent, execute ARA-O, else

Execute appropriate abort mode (select ATO, not AOA-R if already in orbit)

Upper stage decoupler failure

If repairs are possible

Continue burn to ATO

Repair decoupler in orbit

Proceed with mission

If repairs are not possible

Execute appropriate default abort mode for situation (ASTE, AOA, ATO)

Once re-entry guaranteed

HEAT SHIELD MANUAL JETTISON

If unsuccessful, execute ARA-O

In orbit

This section deals with how to react to different failures in orbit. Most of them involve executing a retro burn and returning to Kerbin. Immediate re-entry calls for an immediate retro boost. Normal re-entry is a regular retro boost at a convenient time. Landing point can be selected in this mode. Skip re-entry is a short retro boost so that periapsis will intersect the atmosphere at an altitude no greater than 65 km.

Module	System failure	Abort mode	Notes	Check below
Capsule	Reaction wheel	Immediate re-entry	Use booster gimbal to achieve re-entry attitude	No
	Battery	Immediate re-entry		No
	Food leak	NOT PRESENT		No
	Water leak	NOT PRESENT		No
	O2 leak	Normal re-entry		No
	Computer Battery	Immediate re-entry	Tourism only	No
	Radio x2	Normal re-entry	Pilot decision	No
	Radio	Immediate re-entry	Tourism only	No
	Parachute x1	Normal re-entry	Depending on chance of second parachute failure	No
	Parachute x2	ARA-O		No
SM-1	Engine	Wait for rescue		No
	Engine coolant	Skip re-entry		No
	Engine gimbal	Immediate re-entry		No
	Fins	N/A		No
	LFOX leak	Immediate re-entry		No
	Parachute x1	Ignore		No
	Parachute x2	Ignore	SM-1 will be destroyed on landing	No
	Decoupler	Immediate re-entry ARA-O	ARA-O if heat shield does not detach after burn	No

After re-entry burn

This table shows reactions to system failures after retro burn but before service module detachment.

Module	System failure	Abort mode	Notes	Check below
Capsule	Reaction wheel	Immediately stage		No
	Battery	Immediately stage		No
	Food leak	NOT PRESENT		No
	Water leak	NOT PRESENT		No
	O2 leak	Normal re-entry		No
	Computer Battery	Immediately stage	Tourism only	No
	Radio x2	Normal re-entry	Pilot decision	No
	Radio	Immediately stage	Tourism only	No
	Parachute x1	Normal re-entry	Depending on chance of second parachute failure	No
	Parachute x2	ARA-O		No
SM-1	Engine	Normal re-entry		No

Module	System failure	Abort mode	Notes	Check below
	Engine coolant	Normal re-entry		No
	Engine gimbal	Normal re-entry		No
	Fins	N/A		No
	LFOX leak	Normal re-entry		No
	Parachute x1	Ignore		No
	Parachute x2	Ignore	SM-1 will be destroyed on landing	No
	Decoupler	ARA-O	Attempt to jettison heat shield first	No

During re-entry

Module	System failure	Abort mode	Notes	Check below
Capsule	Reaction wheel	Ignore		No
	Battery	Ignore		No
	Food leak	NOT PRESENT		No
	Water leak	NOT PRESENT		No
	O2 leak	Ignore		No
	Computer Battery	Ignore		No
	Radio x2	Ignore		No
	Radio	Ignore		No
	Parachute x1	Ignore		No
	Parachute x2	CPRE		No